IJCSPUB.ORG ISSN : 2250-1770

INTERNATIONAL JOURNAL OF CURRENT SCIENCE (IJCSPUB)

IJCS PUBLICATION (IJCSPUB.ORG) An International Open Access, Peer-reviewed, Refereed Journal

"Comparative Analysis Of Different Machine Learning Algorithms For Predicting The Results Of International Test Cricket Matches Of India"

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Abstract: This research focuses on the development and comparative analysis of machine learning models for predicting the results of international test cricket matches of India. The study uses logistic regression, decision tree, and random forest algorithms to analyze various features related to teams, players, and match venues. The dataset used for this study includes historical data from international test cricket matches involving India. Feature engineering techniques were used to prepare the dataset for model training and testing. Evaluation of the models was done using various metrics, including accuracy, precision, recall, and F1-score. The results show that the random forest algorithm outperforms the other two algorithms, achieving an accuracy of 85.10%. The study concludes that machine learning algorithms can effectively predict the outcome of international test cricket matches and can be a valuable tool for coaches, analysts, and investors.

Keywords: Logistic Regression, Random Forest, Decision Tree, Cricket Prediction

I. INTROUDUCTION

Cricket is a popular sport worldwide, and its international test matches garner a significant amount of attention from fans, media, and cricket enthusiasts. The outcomes of these matches are determined by various factors, such as the players' skills, team composition, pitch conditions, and weather, making it challenging to predict the result of any given match accurately. However, the development of machine learning algorithms has opened new avenues for predicting the results of cricket matches, including international test matches of India.

In recent years, several researchers have attempted to develop prediction models for cricket matches using machine learning algorithms. These models utilize various data sources, such as player statistics, historical match data, and social media sentiment, to forecast the results of matches. Among the different machine learning algorithms, logistic regression, decision tree, and random forest are popular choices due to their ease of implementation, interpretability, and scalability.

This research paper aims to conduct a comparative analysis of these three machine learning algorithms to predict the results of international test cricket matches of India. The logistic regression algorithm is a linear model that predicts the probability of a binary outcome based on a set of input features. The decision tree algorithm is a non-parametric model that recursively partitions the input space to produce a tree-like

model. The random forest algorithm is an ensemble model that combines multiple decision trees to improve the predictive power.

The purpose of this research is to evaluate the effectiveness of these algorithms for predicting the outcomes of international test cricket matches of India. We will analyze the accuracy, precision, recall, and other performance metrics of each algorithm and compare their strengths and weaknesses. By conducting this comparative analysis, we aim to provide insights into the most effective machine learning algorithm for predicting the results of international test cricket matches of India.

This research paper is structured as follows. In the next section, we provide a literature review of existing research on predicting the results of cricket matches using machine learning. Section III describes the data collection and preprocessing steps. Section IV details the methodology used for the comparative analysis of the machine learning algorithms. Section V presents the results of the analysis, and Section VI discusses the implications of the findings. Finally, we conclude with a summary of the key findings and suggest avenues for future research.

Ultimately, this research aims to provide insights into the effectiveness of different machine learning algorithms for predicting the results of international test cricket matches of India. The findings of this study have practical implications for cricket enthusiasts, analysts, and stakeholders who are interested in predicting the outcomes of cricket matches.

1.1 Importance of study

The study on a comparative analysis of different machine learning algorithms for predicting the results of international test cricket matches of India is important for several reasons:

- Improving the accuracy of cricket match predictions: By comparing the effectiveness of different machine learning algorithms, this study can help identify the most accurate algorithm for predicting the results of international test cricket matches of India. This, in turn, can lead to more accurate predictions and better decision-making for cricket enthusiasts, analysts, and stakeholders.
- Enhancing the understanding of machine learning algorithms: This study can help enhance the understanding of machine learning algorithms, including logistic regression, decision tree, and random forest. By examining the strengths and weaknesses of each algorithm in the context of predicting the outcomes of cricket matches, researchers can gain insights into how these algorithms work and how they can be improved.
- Contributing to the field of sports analytics: The application of machine learning algorithms in predicting the outcomes of cricket matches is an emerging field of sports analytics. This study can contribute to the development of this field by providing insights into the most effective algorithms and highlighting areas for future research.
- Providing insights for cricket stakeholders: The results of this study can provide valuable insights for cricket stakeholders, including coaches, players, team owners, and media professionals. By understanding which factors contribute to successful predictions, stakeholders can make better decisions related to player selection, team composition, and match strategies.
- Advancing the field of machine learning research: By evaluating the effectiveness of different machine learning algorithms, this study can contribute to the development of the field of machine learning research. By identifying the most effective algorithm for predicting the results of cricket matches, researchers can gain insights into how machine learning algorithms can be improved and applied to other domains.

The study on a comparative analysis of different machine learning algorithms for predicting the results of international test cricket matches of India is important for improving the accuracy of cricket match predictions, enhancing the understanding of machine learning algorithms, contributing to the field of sports analytics, providing insights for cricket stakeholders, and advancing the field of machine learning research.

II. LITERATURE REVIEW

C. Troussas et al. [2015] This research focuses on the application of data clustering and mining techniques in the education arena, specifically in modeling student performance. The study aims to compare the effectiveness of four classification algorithms - k-means algorithm, k-Nearest Neighbors algorithm, Support Vector Machines, and Naive Bayes Classifier - in predicting student characteristics. The research recognizes the importance of accurate student performance modeling for educators and students in order to make necessary adjustments to the curriculum and teaching methods. The study utilized several classification and regression models to analyze and compare the data mining results. While previous research has been conducted on algorithmic performance in student modeling, the study aims to draw a concrete conclusion on the similarities and differences between the chosen algorithms. The findings of this study can provide valuable insights to educators and students in terms of better understanding student characteristics and improving the learning process.[1]

S. Priya et al [2022] The research focuses on the use of machine learning algorithms for predicting the winning team in cricket matches. It highlights the difficulty in accurately forecasting the winner of cricket matches and how machine learning has improved prediction power. The study aims to perform a comparative analysis of various supervised classification algorithms such as logistic regression, random forest regression, k-Nearest Neighbor, Support Vector Machine, Naïve Bayes, and Decision Tree to determine their effectiveness in predicting the winning team. The research recognizes that cricket is a popular sport globally and particularly in India, and cricket predictions have always been of interest to fans. The paper highlights that individual players, teams, games, and seasons in sports provide a variety of statistical data that can be utilized in machine learning algorithms for better predictions. The findings of this research can be used to improve the accuracy of cricket predictions and enhance the fan experience. [2]

M. M. Hatharasinghe et al [2019] This research explores the promising area of using computer intelligence to analyze and model the game of Cricket. Cricket's increased popularity and financial benefits have made it an interesting sport for statistical analysis and machine learning. However, the complex rules and dynamic nature of Cricket present challenges for modeling and prediction. The approaches taken in previous research have not been properly documented or analyzed, leading to gaps in the existing knowledge. This paper aims to address these gaps by analyzing the existing work related to match outcome prediction in the Cricket domain. The research examines the strengths and limitations of the previous approaches to inform future research in this area. The paper is the result of an ongoing research project, and the authors hope to fill the missing links and address the drawbacks identified in their analysis. By providing a detailed review of existing research, this paper can contribute to the development of more effective methods for predicting the outcome of Cricket matches using computer intelligence. [3]

M. A. Pramanik et al. [2022] This study examines the importance of cricket as a popular sport and the growth of businesses around cricketing events. The paper highlights the numerous research studies that have been conducted to predict the winner of cricket matches or analyze game statistics to help investors, franchisees, coaches, sports analysts, and technicians make informed decisions. The authors then present their own study, which involves a comparative analysis of different non-ensemble and ensemble machine learning classifiers. The data collected for the study includes all seasons of the Bangladesh Premier League (BPL) T20 tournament, with two datasets prepared: one containing pre-match information and the other including post-match information. The authors applied five base classifiers and five ensemble classifiers for the analysis and found that the K-Nearest Neighbor (KNN) algorithm performed best compared to other

algorithms when predicting match results before the game starts. However, the Gradient Boosting classifier was more robust than other classifiers when considering all features to predict match outcomes. This research provides valuable insights into the effectiveness of different machine learning algorithms for predicting cricket match outcomes and can inform decision-making for investors, coaches, and analysts in the cricketing industry. [4]

III. RESEARCH METHODOLOGY

The methodology for a research paper on "Comparative Analysis of Different Machine Learning Algorithms for Predicting the Results of International Test Cricket Matches of India" is discussed into the following sections:

3.1 Data Collection: The first step is to collect relevant data for the research study. The data can be sourced from espncrickinfo or statistical databases. The data should include information on the teams, match location, toss won, toss decision, past performances, and other relevant factors that may influence the outcome of the game.

3.2 Data Preprocessing: Once the data has been collected, it needs to be preprocessed to ensure that it is clean and ready for analysis. This includes removing any duplicates, missing values, or outliers, and converting the data into a suitable format for analysis.[5]

3.2.1 Exploratory Data Analysis (EDA) : EDA is an important step in any data-driven research project. It helps to gain insights into the data, identify patterns, anomalies, and outliers, and formulate hypotheses that can be tested through statistical models.[8]

3.2.1.1 Univariate Analysis: Univariate Exploratory analysis is use to understand the single variable present in the dataset. We can check distribution, density of particular variable by using different visualization techniques such as countplot, histogram. Some univariate analysis is given in following figures.

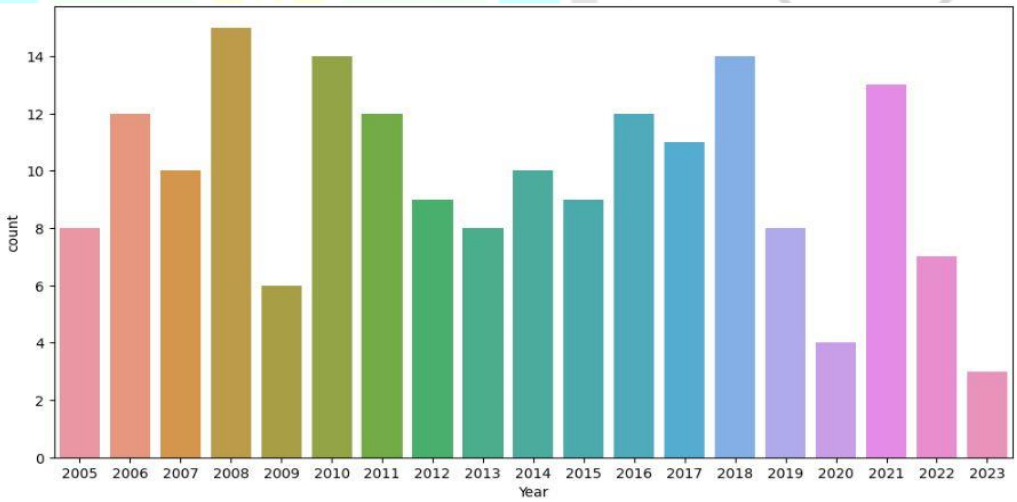

Figure 1. Univariate analysis of variable“Year” using countplot.

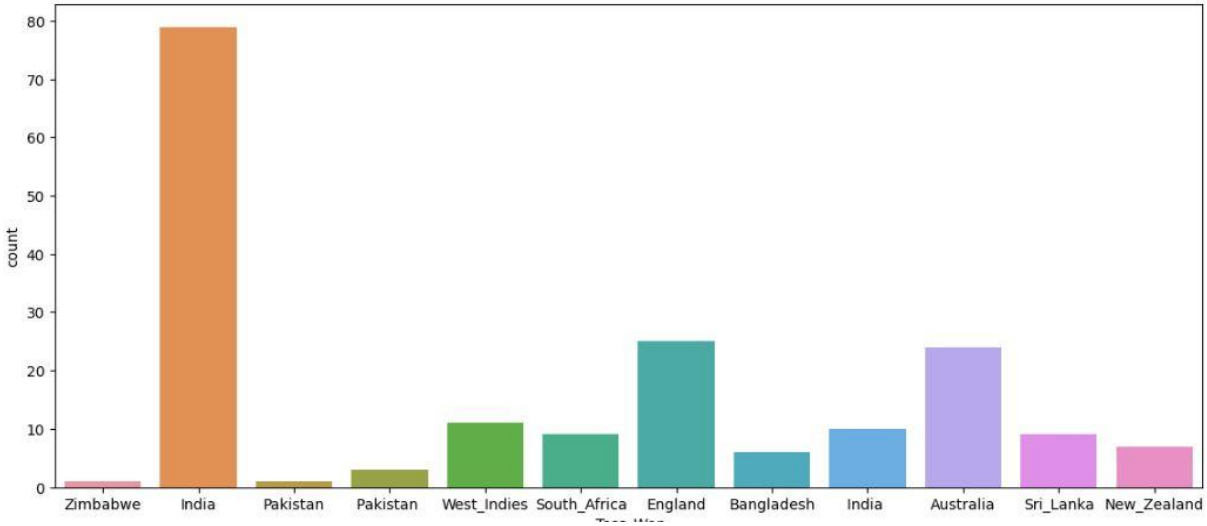

Figure 2. Univariate Analysis of a Variable"Toss_Won" using countplot.

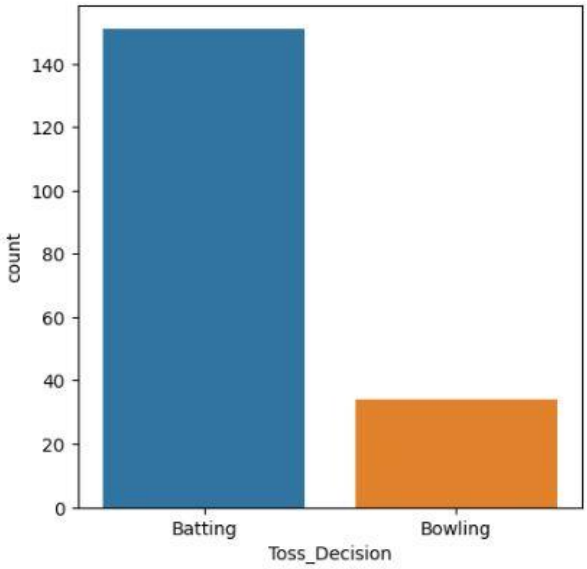

Figure 3. Univariate analysis of a variable"Toss_Decision" using countplot.

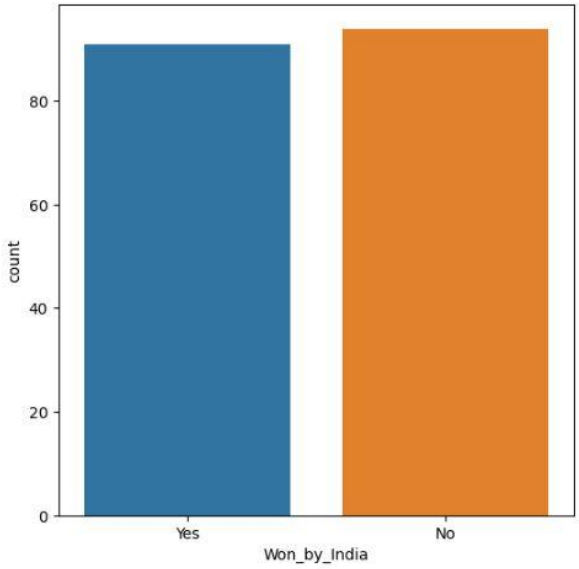

Figure 4. Univariate analysis of a target variable "Won_by_India" using countplot.

3.2.1.2 Multivariate Analysis: Multivariate analysis is a statistical technique used to analyze data that involves multiple variables or factors. It involves the use of mathematical models and statistical algorithms to identify patterns and relationships between multiple variables.[6]

In multivariate analysis, the relationship between multiple independent variables and a dependent variable is studied simultaneously. This enables researchers to examine the complex interactions between variables and to identify the underlying factors that influence the outcome of interest. As we know that, multivariate analysis is the process of finding relationship between variables. In this research we use correlation matrix in order to find relationship between different variables. Similarly visualization of multivariate analysis is shown in the form of Heatmap.[7]

	Match_No	Home_Team	Away_Team	Year	Match_Venue	Toss_Decision	Result_Available	Won_by_India	Toss_Won
Match_No	1.000000	-0.185618	-0.037931	0.997290	0.094648	-0.004181	-0.247990	0.181810	-0.122732
Home_Team	-0.185618	1.000000	-0.557745	-0.145557	-0.071600	0.301318	0.080808	-0.140772	0.325672
Away_Team	-0.037931	-0.557745	1.000000	-0.066775	-0.104861	-0.256740	0.031405	0.237622	0.039723
Year	0.997290	-0.145557	-0.066775	1.000000	0.089466	0.003558	-0.242115	0.171503	-0.118856
Match_Venue	0.094648	-0.071600	-0.104861	0.089466	1.000000	0.090145	0.077972	-0.040883	-0.061866
Toss_Decision	-0.004181	0.301318	-0.256740	0.003558	0.090145	1.000000	-0.079239	-0.048137	0.098553
Result_Available	-0.247990	0.080808	0.031405	-0.242115	0.077972	-0.079239	1.000000	-0.566015	0.036598
Won_by_India	0.181810	-0.140772	0.237622	0.171503	-0.040883	-0.048137	-0.566015	1.000000	-0.009224
Toss_Won	-0.122732	0.325672	0.039723	-0.118856	-0.061866	0.098553	0.036598	-0.009224	1.000000

Figure 5. Multivariate analysis using correlation matrix.

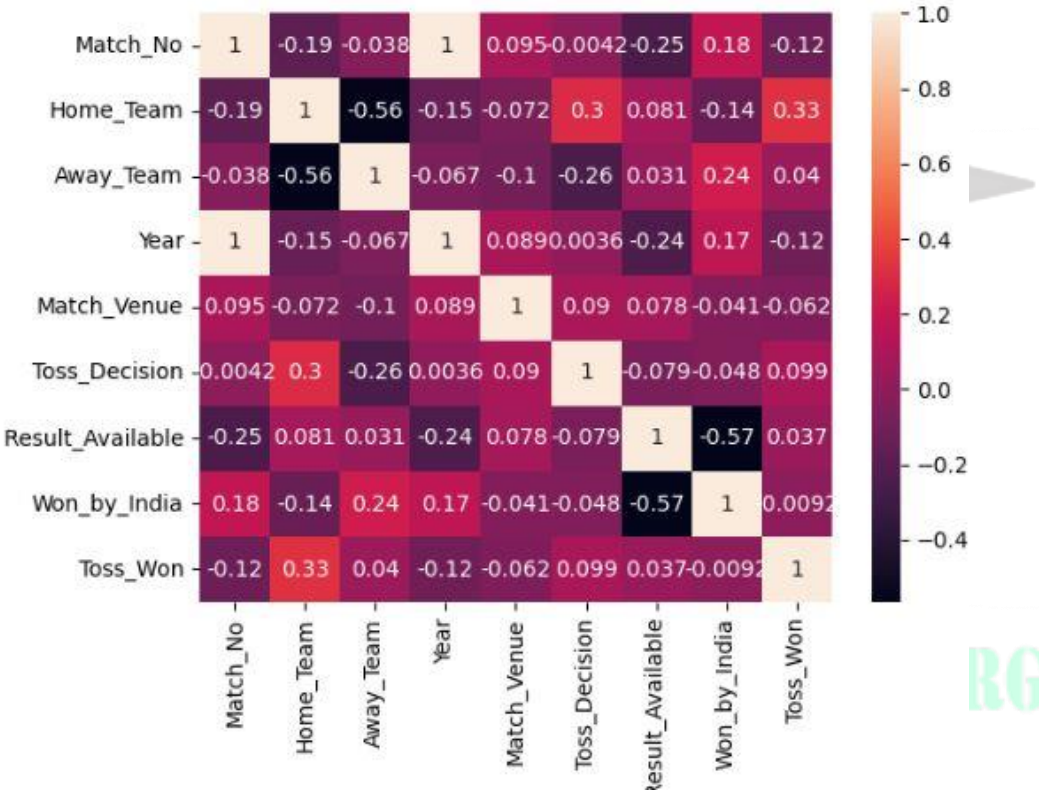

Figure 6. Multivariate Analysis using Heatmap

3.2.2 Feature Encoding: Feature encoding is a crucial step in the data preprocessing phase of machine learning. It involves the conversion of categorical data into numerical data, which can be used as input for machine learning models such as logistic regression, decision trees, and random forest.

In the context of predicting the results of international test cricket matches of India, feature encoding involves converting categorical variables such as the name of the teams, the location of the match, and the type of pitch into numerical data. One common technique for feature encoding is one-hot encoding, which involves creating a binary column for each category and assigning a 1 or 0 for each row depending on whether that category is present in that row.

For example, if the categorical variable is the location of the match and there are three possible locations: Mumbai, Delhi, and Kolkata, then one-hot encoding would create three binary columns, one for each location. If a match was held in Mumbai, then the Mumbai column would have a value of 1 and the Delhi and Kolkata columns would have a value of 0.

Another technique for feature encoding is label encoding, which involves assigning a unique numerical value to each category. For example, if the categorical variable is the type of pitch and there are three possible types: fast, medium, and slow, then label encoding would assign the values 1, 2, and 3 to each type, respectively.

The choice of feature encoding technique depends on the nature of the data and the machine learning algorithm being used. In the case of predicting the results of international test cricket matches of India, one-hot encoding may be a more appropriate technique as it creates binary columns that can be used directly by logistic regression, decision trees, and random forest models.

	Match_No	Home_Team	Away_Team	Match_Venue	Year	Toss_Won	Toss_Decision	Result_Available	Won_by_India
0	1	Zimbabwe	India	Bulawayo	2005	Zimbabwe	Batting	Available	Yes
1	2	Zimbabwe	India	Harare	2005	India	Bowling	Available	Yes
2	3	India	Pakistan	Mohali	2005	India	Batting	drawn	No
3	4	India	Pakistan	Kolkata	2005	India	Batting	Available	Yes
4	5	India	Pakistan	Bengaluru	2005	Pakistan	Batting	Available	No

Figure 7. View of dataframe before feature encoding

	Match_No	Home_Team	Away_Team	Year	Match_Venue	Toss_Decision	Result_Available	Won_by_India	Toss_Won
0	1	10	1	2005	8	0	0	1	10
1	2	10	1	2005	25	1	0	1	1
2	3	1	9	2005	39	0	1	0	1
3	4	1	9	2005	32	0	0	1	1
4	5	1	9	2005	4	0	0	0	9

Figure 8. View of dataframe after feature encoding

3.3 Model Selection:

Logistic Regression, Random Forest, and Decision Tree are the three classification algorithms used in this research paper to predict the results of international test cricket matches of India. Here is an overview of each algorithm and its role in the research:

3.3.1 Logistic Regression: Logistic Regression is a popular classification algorithm that is used to predict the probability of a binary outcome based on input variables. In this research, logistic regression is used to predict the outcome of a cricket match, i.e., whether India will win or lose the match. The algorithm takes in various input variables such as team composition, past performance, pitch conditions, etc., and predicts the outcome based on the logistic function. The logistic function is a mathematical function that maps the input variables to a probability score between 0 and 1.

3.3.2 Random Forest: Random Forest is an ensemble learning technique that combines multiple decision trees to improve the accuracy of predictions. In this research, random forest is used to predict the outcome of a cricket match by creating multiple decision trees from randomly selected subsets of the data. The algorithm then combines the results of each decision tree to make the final prediction. Random forest is preferred over single decision trees because it reduces the risk of overfitting and improves the accuracy of predictions.

3.3.3 Decision Tree: Decision Tree is a popular classification algorithm that is used to create a model that predicts the value of a target variable based on input variables. In this research, decision tree is used to predict the outcome of a cricket match by creating a tree-like model that represents all possible outcomes of the match based on various input variables. The algorithm then traverses the tree to make the final prediction based on the input variables. Decision tree is preferred over other algorithms because it is easy to interpret and provides insights into the decision-making process.

In summary, Logistic Regression, Random Forest, and Decision Tree are the three classification algorithms used in this research to predict the results of international test cricket matches of India. Each algorithm plays a crucial role in the analysis, and their comparative performance is evaluated to determine the most effective algorithm for predicting the outcomes of cricket matches.

3.4. Model Training and Evaluation: The selected algorithms will be trained on the preprocessed data using suitable parameters. The performance of the algorithms will be evaluated using suitable evaluation metrics such as accuracy, precision, recall, and F1 score. The evaluation of the models will be performed using cross-validation to ensure the reliability of the results.

After training the prediction model using various machine learning algorithms, it is essential to evaluate the performance of the model. In this research, we will evaluate the model using different evaluation metrics. These evaluation metrics will help us understand the model's performance and its ability to predict the winning team accurately.

The following are some of the evaluation metrics that will be used in this research:

3.4.1. Accuracy: This evaluation metric measures the percentage of correct predictions made by the model. It is calculated by dividing the number of correct predictions by the total number of predictions. Accuracy score of model is calculated using following formula.

$$\frac{TP + TN}{TP + FP + TN + FN}$$

3.4.2. Precision: This evaluation metric measures the proportion of true positive predictions over the total number of positive predictions made by the model. It is calculated by dividing the number of true positives by the sum of true positives and false positives.

$$\frac{TP}{TP + FP}$$

3.4.3. Recall: This evaluation metric measures the proportion of true positive predictions over the total number of actual positive instances in the dataset. It is calculated by dividing the number of true positives by the sum of true positives and false negatives.

$$\frac{TP}{TP + FN}$$

3.4.4. F1-score: This evaluation metric is the harmonic mean of precision and recall. It provides a balance between precision and recall and is calculated as

$$\frac{2\ X\ (Precision\ X\ Recall)}{Precision + Recall}$$

IV. RESULT AND DISCUSSION

Table 4.1. Performance metrics of three classification algorithm

Algorithm	Accuracy	Precision	Recall	F1-score
Logistic Regression	80.85%	78.26%	81.81%	80.03%
Random Forest	85.10%	82.60%	86.36%	84.48%
Decision Tree	72.34%	73.68%	63.63	68.65%

The table presents the performance metrics of three machine learning algorithms - Logistic Regression, Random Forest, and Decision Tree - based on their accuracy, precision, recall, and F1-score. The results show that the Random Forest algorithm achieved the highest accuracy score of 85.10%, followed by Logistic Regression with 80.85%, and Decision Tree with 72.34%.

In terms of precision, Random Forest also outperformed the other two algorithms, achieving a precision score of 82.60%, while Logistic Regression had a precision score of 78.26% and Decision Tree had a precision score of 73.68%. This suggests that Random Forest had the ability to correctly identify true positives with a higher degree of certainty.

Regarding recall, Random Forest again performed the best, achieving a recall score of 86.36%, while Logistic Regression and Decision Tree had recall scores of 81.81% and 63.63%, respectively. This indicates that Random Forest was better at correctly identifying positive instances. Finally, the F1-score, which is a harmonic mean of precision and recall, was highest for Random Forest at 84.48%, followed by Logistic Regression at 80.03%, and Decision Tree at 68.65%.

Overall, based on the results, it can be concluded that the Random Forest algorithm is the most effective at predicting the outcome of the task being performed, while Logistic Regression is a close second. The Decision Tree algorithm performed the worst, suggesting that it may not be the best choice for this particular task. However, it is important to note that the choice of algorithm ultimately depends on the specific problem at hand, and further experimentation may be required to identify the most suitable algorithm.

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V. CONCLUSION:

In this comparative analysis of different machine learning algorithms for predicting the results of international test cricket matches of India, three algorithms - Logistic Regression, Random Forest, and Decision Tree - were evaluated based on their accuracy, precision, recall, and F1-score. The results suggest that the Random Forest algorithm performed the best, achieving the highest accuracy, precision, recall, and F1-score. This indicates that it is the most effective algorithm for predicting the outcomes of international test cricket matches of India. Logistic Regression also performed well and can be considered a close second.

The findings of this study are particularly useful for cricket enthusiasts, coaches, and analysts who want to predict the outcomes of test cricket matches involving the Indian cricket team. The study highlights the importance of machine learning algorithms in predicting the results of cricket matches and can help improve the accuracy of such predictions. It is important to note that the choice of algorithm may depend on various factors such as the dataset, feature selection, and the problem at hand. Therefore, further experimentation and analysis may be required to identify the most suitable algorithm for specific scenarios.

Overall, this study demonstrates the effectiveness of machine learning algorithms in predicting the results of international test cricket matches and provides valuable insights for future research in this field.

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